Suggested problems: 1-9 odd, 13-25 odd, 31-47 odd

Problem 1: Suppose $\vec{F}(x, y, z) = \langle 4, 0, 5\rangle$ and S is the square of side length 3 perpendicular to the z-axis, centered at $(0, 0, -2)$, and oriented toward the origin. Compute the flux of $\vec{F}$ through S.

We have that $\vec{n} = \vec{k}$, and the area of the square S is 9. Hence the flux is

$$(\vec{F} \cdot \vec{n})(\text{area of } S) = (\langle 4, 0, 5\rangle \cdot \langle 0, 0, 1\rangle)9 = 45.$$

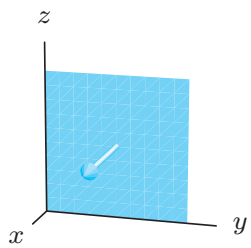
Problem 2: Suppose $\vec{F}(x, y, z) = \langle 2, 0, 3\rangle$ and S is the disk of radius 4 perpendicular to the x-axis, centered at $(5, 0, 0)$, and oriented toward the origin. Compute the flux of $\vec{F}$ through S.

We have that $\vec{n} = -\vec{i}$, and the area of the disk S is 16π. Hence the flux is

$$(\vec{F} \cdot \vec{n})(\text{area of } S) = (\langle 2, 0, 3\rangle \cdot \langle -1, 0, 0\rangle)16\pi = -32\pi.$$

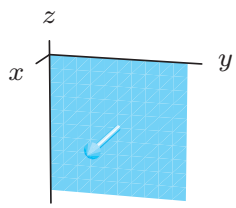
Problem 3: Let $\vec{F}(x, y, z) = \langle z, 0, 0 \rangle$. For each of the oriented surfaces below, decide whether the flux of $\vec{F}$ is positive, negative, or zero.

(a)



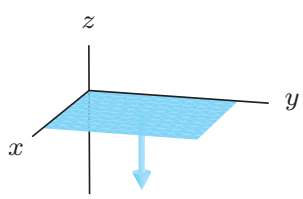
positive

(b)



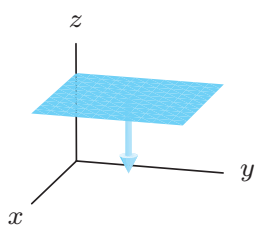
negative

(c)



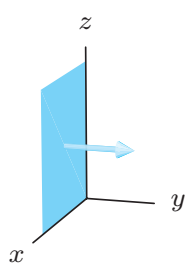
zero

(d)



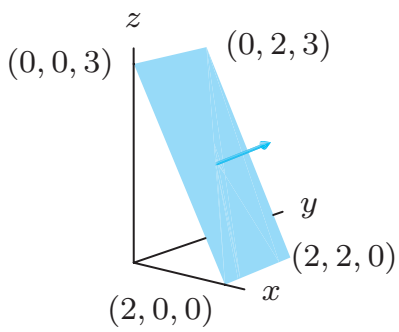
zero

(e)



zero

Problem 4: Suppose $\vec{F}(x, y, z) = \langle 1, 2, -3\rangle$ and that S is the oriented surface below. Compute the flux of F through S.



We have

$$\vec{n} = \frac{1}{\sqrt{13}}\langle 3, 0, 2\rangle.$$

The surface is a rectangle of side lengths 2 and $\sqrt{13}$. Hence

$$(\vec{F} \cdot \vec{n})(\text{area of } S) = \left(\langle 1, 2, -3\rangle \cdot \frac{1}{\sqrt{13}}\langle 3, 0, 2\rangle\right) \cdot 2\sqrt{13} = -6.$$

Problem 5: Let $\vec{F}(x, y, z) = \langle 1, -2, z\rangle$, and suppose S is the part of the graph of $f(x, y) = xy$ lying above the region D which is $0 \le x \le 10$ and $0 \le y \le 10$. Compute $\int_S \vec{F} \cdot d\vec{A}$.

We have

$$\int_S \vec{F} \cdot d\vec{A} = \int_0^{10} \int_0^{10} \langle 1, -2, xy\rangle \cdot \langle -y, -x, 1\rangle \, dxdy = 3000.$$

Problem 6: Let $\vec{F}(x, y, z) = \langle x, 0, z\rangle$, and suppose S is the part of the graph of $f(x, y) = x+y+2$ lying above the region D which is the triangle with vertices $(0, 0, 0)$, $(1, 0, 0)$, and $(0, 1, 0)$. Compute $\int_S \vec{F} \cdot d\vec{A}$.

We have

$$\int_S \vec{F} \cdot d\vec{A} = \int_0^1 \int_0^{-x+1} \langle x, 0, x + y + 2\rangle \cdot \langle -1, -1, 1\rangle \, dydx = \frac{7}{6}.$$